Desk Buddy

Physical controls. Configurable software.

I built Desk Buddy as a programmable desktop controller using commercial modules. Six buttons, a push encoder and a touch display connect to a Windows companion app for profiles, keyboard shortcuts, media control and animated display themes.



Current enclosed prototype. Its internal USB lead exits through a battery-side opening; the computer end remains accessible.

BUILT PROTOTYPE

The printed R6 enclosure is assembled and closed. The display is powered in the current front photograph.

6
physical buttons

12
bindings per profile

3
configurable profiles

Platform
nRF52840
Arduino C++
C# / WPF companion app
USB + Bluetooth LE

Software baseline: Firmware R3H / Hub 1.7.1. **Mechanical milestone:** R6 printed and assembled, with a USB routing workaround.

Historical bench verification is summarized separately; enclosed reliability testing remains next work.

Engineering the interactions

I separated user intent, live readiness and saved configuration so that navigation, connection changes and editing could be handled independently.



Windows Hub manages configuration over USB CDC or authorized, encrypted BLE GATT. Management and host keyboard/media output use separate state and permissions.

01 Keep intent separate from execution

Opening menus or changing appearance had switched output OFF. I separated the requested output state from transmission permission. Navigation now preserves the choice; diagnostics and display sleep suppress sending, and waking does not silently enable output.

02 Measure responsiveness across tasks

Language saves still delayed touch sampling after drawing became cooperative. I changed task priorities and explicit yielding; the sampled maximum polling gap fell from **409 ms to 126 ms** between intermediate and final R3F scoped board runs. This measures polling, not end-to-end touch latency.

03 Make offline state intelligible

Reopening a disconnected Hub showed defaults, while the device retained its settings. I added separate last-read snapshots and unsent drafts, labelled their source, and made fresh readback authoritative. Close/reopen and recovery passed across eight isolated WPF processes with simulated transport.

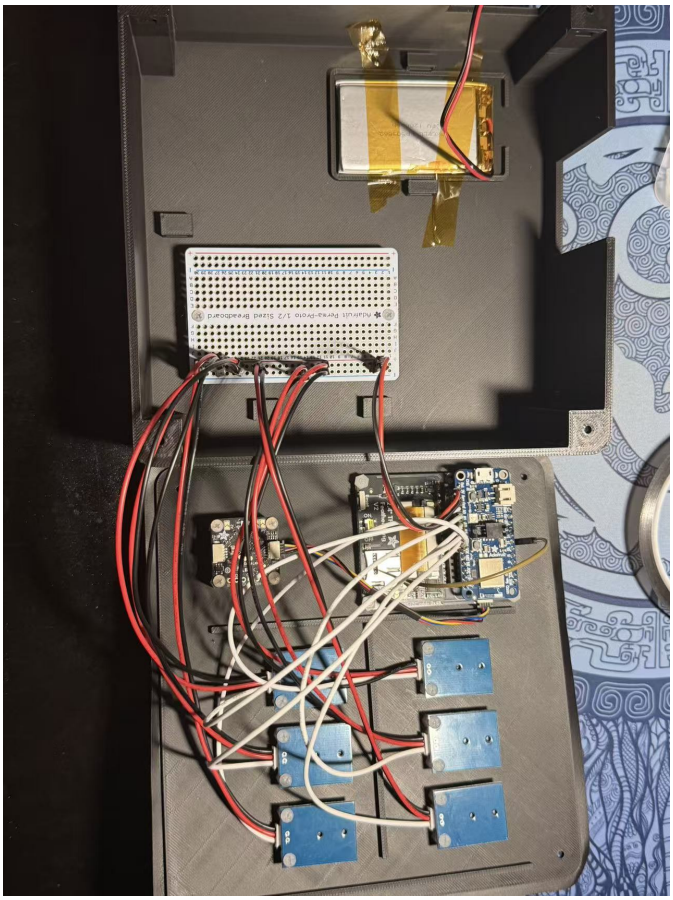
Historical bench evidence

Device: R3H passed real-device USB configuration/readback checks, BLE management and a 30-second automatic display-sleep check. Backlight-off and first-touch wake were physically observed on R3G under USB power. **Desktop:** Hub 1.7.1 passed host regression and offline WPF/lifecycle checks.

These results predate final enclosure assembly. They establish specific software and bench behaviors, not a new enclosed-system qualification. Battery endurance has not been measured.

From CAD to an enclosed build

I used a two-level enclosure: the display, encoder and six buttons mount to the lid; the battery and Perma-Proto board occupy the base, with space for the existing wiring.



Earlier internal assembly, before the final 180-degree display rotation and USB reroute. The final internal cable arrangement was not photographed.

The fit issue

The controller board sat on the right of the display, rather than the left assumed in CAD. As a result, the printed access opening did not expose the USB port.

The working adaptation

I rotated the display assembly 180 degrees, connected USB inside the enclosure, and routed the cable out through the battery-side opening. I now disconnect it at the computer end.

What I would change next

I would correct the board orientation in CAD, check the complete connector and cable-bend envelope against the real assembly, and add a defined strain-relief feature. The published R6 geometry has not yet been revised.

As-built front-panel layout

Viewed from the front, with the screen text upright. Physical positions changed; electrical identities and firmware mappings did not.

LEFT COLUMN

Top: K6 / blue
Middle: K5 / black
Bottom: K4 / red

RIGHT COLUMN

Top: K3 / yellow
Middle: K2 / white
Bottom: K1 / blue

Next verification: enclosed controls and reconnect behavior, cable retention and clearance, battery-only wake, and longer-duration operation.